FOUNDATION COURSE

START YOUR PREPARATION FROM BASICS

Prerequisites of Engineering Mathematics

Lecture no. 04

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Recap of previous lecture

(LOGARITHMS)

$$\log_2 8 = 3$$



Topics *to be covered*

(PARTIAL FRACTIONS)



Case I: when Denominator is in the form of Linear factors ⟶

$$\frac{p(x)}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

where, Deg of N^r < Deg of D^r

Case II: when D^r is in the form of Linear repeated Factors; ⟶

$$\frac{p(x)}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$$

Here Deg of N^r < Deg of D^r

Case III → when D^r is in the form of Quadratic expression :— ⟶

$$\frac{p(x)}{(x-a)(\alpha x^2+\beta x+\gamma)} = \frac{A}{x-a} + \frac{Bx+C}{\alpha x^2+\beta x+\gamma}$$

Here Deg of N^r < Deg of D^r

Q. Resolve following expressions into it's partial fractions:

(1) $\frac{12+9x}{x(x+4)} = ? = \frac{A}{x} + \frac{B}{x+4}$ ——— (1)

$12+9x = A(x+4) + B(x)$ ——— (2)

Put $x=0 \Rightarrow 12+0 = A(4)+0 \Rightarrow A=3$

Again Put $x=-4 \Rightarrow 12-36 = 0-4B \Rightarrow B=6$

So By (1) $\frac{12+9x}{x(x+4)} = \frac{3}{x} + \frac{6}{x+4}$

shortcut:- $\frac{12+0}{(0+4)} = A \Rightarrow A=3$

Again $\frac{12-36}{-4} = B \Rightarrow B=6$

(P8) $\frac{e^x+1}{(e^x+2)(e^x+3)} = ?$ Let $e^x = X$

$\Rightarrow \frac{X+1}{(X+2)(X+3)} = \frac{A}{X+2} + \frac{B}{X+3} = \frac{-1}{e^x+2} + \frac{2}{e^x+3}$

for $X=-2$ $\frac{-2+1}{-2+3} = A \Rightarrow A=-1$

for $X=-3$ $\frac{-3+1}{-3+2} = B \Rightarrow B=2$

Q. $\frac{x+3}{x^3+7x^2+10x} = ? = \frac{x+3}{x(x^2+7x+10)} = \frac{x+3}{x(x+2)(x+5)}$

$$\frac{x+3}{x(x+2)(x+5)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x+5} \quad \text{---(1)}$$

$$x+3 = A(x+2)(x+5) + B(x)(x+5) + C\,x(x+2) \quad \text{---(2)}$$

Put $x=-2$, $-2+3 = 0 + B(-2)(3) + 0 \Rightarrow B = -1/6$

Put $x=-5$, $-5+3 = 0 + 0 + C(-5)(-3) \Rightarrow C = \frac{-2}{15}$

Put $x=0$, $0+3 = A(2)(5) + 0 + 0 \Rightarrow A = 3/10$

Soln By (1), $\frac{x+3}{x^3+7x^2+10x} = \frac{3/10}{x} + \frac{-1/6}{x+2} + \frac{-2/15}{x+5}$

M-II $\frac{x+3}{x(x+2)(x+5)} = \frac{A}{x} + \frac{B}{x+2} + \frac{C}{x+5}$

for $x=0$, $\frac{3}{(2)(5)} = A \Rightarrow A = \frac{3}{10}$

for $x=-2$, $\frac{1}{(-2)(3)} = B \Rightarrow B = \frac{-1}{6}$

for $x=-5$, $\frac{-2}{(-5)(-3)} = C \Rightarrow C = \frac{-2}{15}$

Q. $\frac{1}{x^3+x^2} = ? = \frac{1}{x^2(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x+1}$ ——— (1)

i.e. $1 = A\,x(x+1) + B(x+1) + C(x^2)$ ——— (2)

Put $x=0$, $1 = 0 + B + 0 \Rightarrow B = 1$

Put $x=-1$, $1 = 0 + 0 + C \Rightarrow C = 1$

Put $x=1$, $1 = 2A + 2B + C$

$1 = 2A + 2 + 1 \Rightarrow A = -1$

So $\frac{1}{x^3+x^2} = \frac{-1}{x} + \frac{1}{x^2} + \frac{1}{x+1}$

Q. $\frac{9}{(x-1)(x+2)^2} = ? = \frac{A}{(x-1)} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2}$ ——— (1)

$9 = A(x+2)^2 + B(x-1)(x+2) + C(x-1)$ ——— (2)

Put $x=-2$, $9 = 0 + 0 + C(-2-1) \Rightarrow C = -3$

Put $x=1$, $9 = A(1+2)^2 + 0 + 0 \Rightarrow A = 1$

Put $x=0$, $9 = 4A - 2B - C$

$9 = 4 - 2B + 3 \Rightarrow B = -1$

Q. $\frac{x^2+23x}{(x+3)(x^2+1)} = ? = \frac{A}{x+3} + \frac{Bx+C}{x^2+1}$ ——— (1)

$\Rightarrow x^2+23x = A(x^2+1)+(Bx+C)(x+3)$ ——— (2)

Put $x=-3$, $9+23(-3) = A(9+1)+0 \Rightarrow A = \frac{9-69}{10} = -6$

Put $x=0$, $0 = A(1)+(0+C)(3) \Rightarrow A+3C=0 \Rightarrow C = \frac{-A}{3} = 2$

Put $x=1$, $1+23 = 2A+(B+C)(1+3) \Rightarrow 24 = -12+4B+4C$

$24 = -12+4B+8$

$4B = 28 \Rightarrow B = 7$

$\Rightarrow \frac{x^2+23x}{(x+3)(x^2+1)} = \frac{-6}{x+3} + \frac{7x+2}{x^2+1}$ //

Q. $\frac{(x+4)}{(x-3)(x^2-2)} = ? = \frac{A}{x-3} + \frac{Bx+C}{x^2-2}$ — (1)

$x+4 = A(x^2-2) + (Bx+C)(x-3)$ — (2)

Put $x=3$, $3+4 = A(9-2) + 0 \Rightarrow A = 1$

Put $x=0$, $4 = A(0-2) + (0+C)(-3) \Rightarrow 4 = -2A - 3C \Rightarrow C = \frac{6}{-3} = -2$

Put $x=1$, $1+4 = A(-1) + (B+C)(-2) \Rightarrow 5 = -A - 2B - 2C$

$5 = -1 - 2B + 4 \Rightarrow B = \frac{2}{-2} = -1$

$$\frac{x+4}{(x-3)(x^2-2)} = \frac{1}{x-3} + \frac{-x-2}{x^2-2}$$ //


Thank
YOU